

PHP

- ⊛ PHP is; 'Server Scripting Language' which dynamic and interactive web pages allow users to interact with.
- ⊛ PHP is an acronym for "PHP: Hypertext Preprocessor"
- ⊛ PHP file which text, HTML, CSS, Javascript and PHP codes which are written in ".php" file extension which is required.

- ⊛ PHP syntax structure;

<?PHP

// PHP code goes here

?>

practical.

- ⊛ PHP code which server which is execute and then; web browser which is plain HTML which is output to.
- ⊛ HTML code which is written in PHP codes which are written in (<body> tag)
- ⊛ PHP statement which is written in semicolon (;) which is.
- ⊛ PHP is ; keywords, classes, functions, user-defined functions "case-sensitive" which is. which; variable names "case-sensitive" which is.

Comments

- ⊛ Single line comments which is written in // which is.

// This is comment

This is comment

- ⊛ Multiline comments which is written in /* which is.

/* This is

multiline

comment */

Variables

- ⊛ PHP is variable name which is written in \$ which is.
- ⊛ strings assign which is written in double quotation marks and single quotation marks which is.
- ⊛ variables which is written in \$ which is.
 - 1) variable which is written in dollar sign (\$) which is.
 - 2) variable name which is written in underscore and letter which is.

- 3) variable name should number should not start.
- 4) variable name should alpha numeric characters and underscore should be used.
- 5) variable names, case sensitive.

* variable names output verification; echo keyword used.

```

Ex:- $txt = "Praveen";
      echo "Hello $txt";

```

like this you can use echo keyword.

```

Ex:- $txt = "Praveen";
      echo "Hello ". $txt. "Welcome";

```

* variable should data type should use var_dump() on-build function.

```

Ex:- $x = 5;
      var_dump($x);

```

variable scope

* The scope of a variable is the part of the script where the variable can be ~~referenced~~ referenced / used.

* variable scope types 3 are.

- (i.) local
- (ii.) global
- (iii.) static

* function should declare ~~variables~~ variables, global scope should use. global keyword should use in function.

* function should declare local scope should use. local keyword should use in function.

* * global keyword should use in function to access global variable.

```

Ex:- function myTest() {
      global $x;
    }

```

* PHP we can use global variables, \$GLOBALS[index] where array should use. index should use variable name should use in function.

```

Ex:- function myTest() {
      $GLOBALS['x'];
    }

```



3

Echo / Print

- echo

- 27

print

- ??

Data types

① PHP ରେ ୫; string, Integer, float (floating point numbers - also called double), Boolean, Array, Object, NULL, Resource ଏହି data types କહેଲେ.

② data type ଯାହା ଭେଦ କରିବା var_dump() on-build function କହାଯାଏ.

③ Integers ରେ ୫; -2, 147, 483, 648 ଏବଂ 2, 147, 483, 647 ମଧ୍ୟରୁ କେବଳ ଧନାତ୍ମକ ସଂଖ୍ୟା ଗଣ୍ୟ ଅଟେ.

④ Integers ; decimal (base 10), hexadecimal (base 16), Octal (base 8) or binary (base 2) notation ଯୋଗାଇ ଦିଆଯାଏ.

⑤ boolean values ଦୁଇ; true ଏବଂ false ଅଟେ.

⑥ values କିଛିତରଙ୍ଗର ଯାହା variable ଯାହା store କରିବେ Arrays କହାଯାଏ.

ଉଦା:- `$cars = array("Volvo", "BMW", "Ford");`

⑦ NULL data type ଯେତେବେଳେ କିଛିତରଙ୍ଗର ଯାହା, null value ଯାହା ଥାଏ. ତେଣୁ values assign କରାଯିବା ନାହିଁ କହାଯାଏ.

ଉଦା:- `$x = null;`
`echo $x;`

⑧ Data type ଯାହା ତେଣୁ କରିବେ କିଛିତରଙ୍ଗର କିଛିତରଙ୍ଗର ମଧ୍ୟରୁ କିଛିତରଙ୍ଗର ମଧ୍ୟରୁ; ତେଣୁ type ଯାହା value ଯାହା assign କରିବେ type casting କହାଯାଏ.

ଉଦା:- `$x = 5;`
`$x = "Hello";`
`var_dump($x); // Outputs string(5) "Hello"`

⑨ ଏବଂ assign କରାଯିବା value ଯାହା ତେଣୁ କିଛିତରଙ୍ଗର type casting କରିବେ କିଛିତରଙ୍ଗର କିଛିତରଙ୍ଗର କହାଯାଏ.

ଉଦା:- `$x = 5;`
`$x = (string)$x;`
`var_dump($x); // Outputs string(1) "5"`

strings

① string ଯାହା length ଯାହା ଦେଖିବା strlen() on-build function ଯାହା କହାଯାଏ.

ଉଦା:- `echo strlen("Hello World!");`

② string ଯାହା ମଧ୍ୟରୁ କିଛିତରଙ୍ଗର (separate from whitespaces) କିଛିତରଙ୍ଗର ~~str_word_count~~ str_word_count() on-build function ଯାହା କହାଯାଏ.

ଉଦା:- `echo str_word_count("Hello World!");`

★ string શોધવા માટે `strpos()` on-build function
કેળે. કોઈ પણ string શોધ કરવામાં આવે, તે string શોધવાની જગ્યા પર
જોઈ index શોધ return કરે. નહીં હોય તો, false return કરે.

જેમ:- `echo strpos("Hello World!", "World!");`

★ string શોધવા uppercase કરવા `strtoupper()` on-build function શોધ, `strtolower()` નીચે lowercase કરવા, બંનેમાંથી પસંદ.

જેમ:- `$x = "Hello";`

`echo strtoupper($x); // HELLO`

`echo strtolower($x); // hello`

★ જો string શોધવામાં આવે string શોધવાની `replace` કરે શકાય. જો કોઈ `str_replace()`
on-build function કેળે.

જેમ:- `$x = "Hello World!";`

`echo str_replace("World", "Dolly", $x);`

★ string શોધવા reverse કરવા `strrev()` on-build function કેળે.

જેમ:- `$x = "Hello World!";`

`echo strrev($x);`

★ string શોધવા જેમાં જોઈ whitespaces હોય તેને `trim()` on-build function
કેળે શકાય.

જેમ:- `$x = "Hello World!";`

`echo trim($x);`

★ string શોધવા Array શોધવા માટે `split` કરવા `explode()` on-build function
શોધ કેળે.

જેમ:- `$x = "Hello World!";`

`$y = explode(" ", $x);`

// Use the `print_r()` function to display the result
`print_r($y);`

/*

Result:

Array ([0] => Hello [1] => World!)

*/

★ strings concatenate કરવા માટે `concat` કરવા.

જેમ:- 1.) `$x = "Hello";`

`$y = "World";`

`$z = $x . $y; // Use dot operator (.)`

`echo $z; // Outputs HelloWorld`

★ જો કોઈ કોઈ
" " નીચે
જો.

2) double space whitespace କ୍ଷେତ୍ର;

```
$2 = $x. " ". $y;
```

```
echo $2; // Outputs Hello World
```

3) dot operator ଶାନ୍ତେ double quotes ମଧ୍ୟ, କ୍ଷେତ୍ର ଥାଏ.

```
$2 = "$x $y";
```

```
echo $2; // Outputs Hello World
```

★ string ଥିବା slice କ୍ଷେତ୍ର substr() on-build function କ୍ଷେତ୍ର.

★ ଯେଉଁ ସ୍ଥଳେ କ୍ଷେତ୍ରରେ slice କ୍ଷେତ୍ର ଥିବାରୁ index ଥିବା ଥିବା ମାନ, ଯେଉଁ କ୍ଷେତ୍ରରେ ଥିବା index ଥିବାରୁ end ମାନ କ୍ଷେତ୍ର ମାନ ଥିବା ଥିବା characters କ୍ଷେତ୍ର.

```
$x = "Hello World!";
```

```
echo substr($x, 6, 5); // Outputs World
```

★ end ଥିବା ଥିବା ମାନେ slice କ୍ଷେତ୍ର ମାନ; ଥିବା argument ଥିବା କ୍ଷେତ୍ର କ୍ଷେତ୍ର.

```
echo substr($x, 6);
```

Escape characters

★ string ଥିବା କ୍ଷେତ୍ରରେ errors କ୍ଷେତ୍ର ଥିବା characters କ୍ଷେତ୍ର escape characters କ୍ଷେତ୍ର.

code	result
'	single quote
"	double quote
\$	PHP variables
\n	New line
\r	Carriage return
\t	Tab space
\f	Form feed
1000	Octal value
1xhh	Hex value

Numbers

★ numeric types ଥିବା ମାନ.

★ Integer

★ Float

★ Number Strings

★ ଥିବା ଥିବା data types ଥିବା ମାନ.

★ Infinity

★ NaN

★ integer වෙනුවෙන් දෘශ්‍යමාන්වලින් ගබඩා වේ.

Note: $4 * 2.5$ is 10, the result is stored as float, because one of the operands is a float (2.5)

★ ඒ variable වෙනුවෙන් data type වන integer හි ඇති බැවින් තවත් on-build functions

`is_int()` , `is_integer()` , `is_long()`

★ ඒ variable වෙනුවෙන් data type වන float හි ඇති බැවින් තවත් on-build functions

`is_float()` , `is_double()`

★ float data-type යටතේ store කළ හැකි විශාලතම සංඛ්‍යාවට වඩා විශාල සංඛ්‍යාවක් variable වෙනුවෙන් assign කළ විට, එය infinity value වෙනුවෙන් ලැබේ.

★ ඒ value වෙනුවෙන් infinite ද, එසේ නොමැති බැවින් තවත් on-build functions

`is_finite()` , `is_infinite()`

★ NaN යන්නෙන්, 'Not a Number' යන්නෙන් හැදින්වේ. impossible mathematical operations වල දී ලැබේ.

★ value වෙනුවෙන් Not a Number හි ඇති බැවින් තවත් `is_nan()` on-build function

★ PHP variable වෙනුවෙන් numerical value වෙනුවෙන් store කර ඇති ඇති බැවින් `is_numeric()` on-build function වන බැවින්, true සහ false output

★ Strings සහ Floats, Integers බවට cast කිරීමට `(int)`, `(integer)`, `intval()` යන ක්‍රම යොදා ගත හැක.

Cast

★ PHP casting වල දී තවත් statements ලියා ඇත.

★ (string) - Converts to data type String

★ (int) - Converts to data type Integer

★ (float) - Converts to data type Float

★ (bool) - Converts to data type Boolean

★ (array) - Converts to data type Array

★ (object) - Converts to data type Object

★ (unset) - Converts to data type NULL

- Boolean value cast කිරීමේදී value එක 0, NULL, false හෝ empty නම්; false ලෙස ද, අනෙකින් true ලෙස ද, output කරයි.

Math

- π ගුණක ලබා ගැනීමට pi() on-build function එක යොදයි.
 උදා:- echo(pi());
- values list එකකින් ඉහත හෝ පහත value එක සෙවීමට min() හෝ max() on-build functions යොදිය හැක.
 උදා:- echo(min(0, 150, 30, 20, -8, -200));
 echo(max(0, 150, 30, 20, -8, -200));
- absolute (positive) value එක ලබා ගැනීමට abs() on-build function එක යොදයි.
 උදා:- echo(abs(-6.7));
- number එකක square root එක ලබා ගැනීමට sqrt() on-build function එක යොදිය හැක.
 උදා:- echo(sqrt(64));
- දශම සංඛ්‍යාවක් ඉරට්ටේ දුරස්ත සංඛ්‍යාව ලබා ගැනීමට round() on-build function එක යොදයි.
 උදා:- echo(round(0.49));
- Random number එකක් generate කිරීමට rand() on-build function එක යොදයි.
 උදා:- echo(rand());
- ★ නිශ්චිතව සංඛ්‍යා පරාසයක දී random number එකක් generate කිරීමට ද, මෙය සිදුකරනු ලබයි. මෙහිදී පහත දැක්වෙන ආකාරයට, ඉහත සිදු කරනු ලබයි.
 උදා:- echo(rand(10, 100));

Constants

- මීට variables ඉවත්කරමින් හෝ ඉවත් කිරීමට define හෝ undefine කිරීමට යොදා ගනී.
- Constants, script එක පුරාම ~~global~~ global scope වේ.
- Constants නිර්වචනය කිරීමට define() on-build function යොදයි. නම syntax යොදයි.

define(name, value, case-insensitive);

parameters;

name ⇒ specifies the name of the constant

value ⇒ specifies the value of the constant

case-insensitive ⇒ specifies ~~the~~ whether the constant name should be case-insensitive. default is false

2201-1.) Create a constant with a case-sensitive name;
define ("GREETING", "Welcome");
echo GREETING;

2.) Create a constant with case-insensitive name;

```
define("GREETING", "Welcome", true);  
echo GREETING;
```

⑤ constants ચિહ્નો const keyword થી બનાવવામાં આવે છે.
 ex: const int x;

```
220: const MYCAR = "Volvo";  
      echo MYCAR;
```

⑤ const અને define() માર્કર છેલ્લેથી ચર્ચિત્વની.

- * `const` are always case-sensitive
- * `define()` has a case-sensitive option
- * `const` cannot be created inside another block scope, like inside a function or inside an `if` statement.
- * `define` can be created inside another block scope

④ define() on-build function ଥିବାର constant Array ଥିବାର ସିରିଆଲାଇଜେସନ୍ ଥିବାର ସୁବିଧା ଥିବାର।

```
2204 define ("cars", ["Alfa Romeo", "BMW", "Toyota"]);
      echo cars [0];
```

Operators

⑤ Arithmetic operators,

operator	Example
+	$\$x + \y
-	$\$x - \y
*	$\$x * \y
/	$\$x / \y
%	$\$x \% \y
**	$\$x ** \y

④ Assignment operators;

operator	same as ...
$x = y$	$x = y$
$x += y$	$x = x + y$
$x -= y$	$x = x - y$
$x *= y$	$x = x * y$
$x /= y$	$x = x / y$
$x \% = y$	$x = x \% y$

④ Comparison operators;

operator	result
$\$x == \y	Returns true if $\$x$ is equal to $\$y$
$\$x === \y	Returns true if $\$x$ is equal to $\$y$, and they are of the same type
$\$x != \y	Returns true if $\$x$ is not equal to $\$y$
$\$x <> \y	Returns true if $\$x$ is not equal to $\$y$
$\$x !== \y	Returns true if $\$x$ is not equal to $\$y$ or they are not of the same type
$\$x > \y	Returns true if $\$x$ is greater than $\$y$
$\$x < \y	Returns true if $\$x$ is lower / less than $\$y$
$\$x >= \y	Returns true if $\$x$ is greater than or equal to $\$y$
$\$x <= \y	Returns true if $\$x$ is less than, or equal to $\$y$
$\$x <=> \y	Returns an integer less than, equal to or greater than zero, depending on if $\$x$ is less than, equal to or greater than $\$y$.

★ Increment / Decrement operators;

operator	description
$++ \$x$	Increments $\$x$ by one, then returns $\$x$
$\$x ++$	Returns $\$x$, then increments $\$x$ by one
$-- \$x$	Decrements $\$x$ by one, then returns $\$x$
$\$x --$	Returns $\$x$, then decrements $\$x$ by one

★ Logical operators;

operator	Example	description
and	$\$x$ and $\$y$	True if both $\$x$ and $\$y$ are true
or	$\$x$ or $\$y$	True if either $\$x$ or $\$y$ is true
xor	$\$x$ xor $\$y$	True if either $\$x$ or $\$y$ is true, but not both
&&	$\$x \&\& \y	True if both $\$x$ and $\$y$ are true
	$\$x \y	True if either $\$x$ or $\$y$ is true
!	! $\$x$	True if $\$x$ is not true

Conditional statements

- ★ condition එකක් සත්‍ය වූයේදැයි, තීරණය කිරීමෙන් පසු නිර්ණය කළ හැකි if statements සෑදිය හැක.

```

උදා: $X = 14;
    if ($X == 14) {
        echo "Have a good day";
    }

```

- ★ condition එකක් සත්‍ය නම්, තීරණය කිරීමෙන් පසු නිර්ණය කළ හැකි; ප්‍රතික්ෂේප නම්, තවත් ක්‍රියාවක් කළ හැකි; if-else statements සෑදිය හැක.

```

උදා: $X = 14;
    if ($X == 14) {
        echo "Hello";
    } else {
        echo "Hi";
    }

```


- ★ **if** statement: **if** condition is true, then the code block under **if** will execute; if condition is false, then the code block under **if** will not execute; **if-else** statements.

```

$X = 4;
if ($X < 10) {
    echo "less than 10";
} elseif (($X < 15)) {
    echo "less than 15";
} else {
    echo "greater than 15";
}

```

- ★ **if** statements: **if** condition is true, then the code block under **if** will execute.

```

$a = 5;
if ($a < 10) $b = "Hello";
echo $b;

```

- ★ **if-else** statements: **if** condition is true, then the code block under **if** will execute; if condition is false, then the code block under **else** will execute.

```

$a = 13;
$b = $a < 10 ? "Hello" : "Good Bye";
echo $b;

```

- ★ **switch** statements: **switch** statement is used to execute one of the many code blocks under **switch** statements.

- * **break** keyword: **break** keyword is used to break the **switch** block when the condition is true.
- * **default** keyword: **default** keyword is used to execute the code block when the condition is false.

```

$favcolor = "red";
switch ($favcolor) {
    case "red":
        echo "Your favourite color is red!";
        break;
    case "blue":
        echo "Your favourite color is blue!";
        break;
    default:
        echo "Your favourite color is neither red, nor blue!";
}

```

- * **break**: **break** keyword is used to break the code blocks execute when the condition is true.

* default: જો switch statement થી કોઈ case code block થી કોઈ case ની કોડ બ્લોક નથી મળે તો default case ની કોડ બ્લોક ચાલે. જો default case ની કોડ બ્લોક નથી તો switch statement ની કોડ બ્લોક નથી ચાલે.

* cases ରେ ବ୍ୟବହାର କରାଯାଇ code block ରେ ଏହା କୋଡ୍ ଭାବରେ ଥାଏ, ଯାହାକି.

ଉଦା:- $d = 3;$

2001 \$d = 3;

```
switch ($?) {
```

Case 1:

Case 2:

case 3 :

Case 4:

case 5:

echo "The weeks feels so long!";
break;

Case 6:

Case 0:

echo "Weekends are the best!";
break;

default:

echo "Something went wrong!";

3

Loops

④ සමී කොරැල්ලිස්සාල යටතේ කිසියම් ක්‍රියාවක් ගැන ගැන සිදු කිරීම 100ps කොටයි.
100ps වර්ග කිසියම් ක්‍රියාව වේ.

- 1.) while - loops through a block of code as long as the specified condition is true
- 2.) do-while - loops through a block of code once, and then repeats as long as the specified condition is true
- 3.) for - loops through a block of code a specified number of times
- 4.) foreach - loops through a block of code for each element in the array

while loop

④ କିଛିଟି 'କୋରଡ଼ିଙ୍ଗ' ଘରର ପୋରା ଗାଈର ଖୁସି ଭାବେ.

ဥပမာ- \$i = 1;

```
while ($i < 6) {  
    echo $i;  
    $i++;  
}
```

* increment operator သည်
အတိတ်ကုန်ဆုံးကို ပြောင်းလဲပေးသည်။
အဆိုပါနည်းလမ်းဖြင့် loop သည် အတိတ်ကုန်ဆုံး
အထိ ဆက်လက်လည်ပတ်သည်။

⊛ break statement သည် loop condition သည် true ဖြစ်သည့်အခါမှ စတင်ပြီး loop သည်
အတိတ်ကုန်ဆုံးသို့ ရောက်ရှိသည်။

ဥပမာ- \$x = 1;

```
while ($x < 6) {  
    if ($x == 3) break;  
    echo $x;  
    $x++;  
}
```

⊛ continue statement သည် loop iteration သည် အတိတ်ကုန်ဆုံးသို့ ရောက်ရှိသည့်အခါမှ စတင်ပြီး
loop သည် အတိတ်ကုန်ဆုံးသို့ ရောက်ရှိသည်။

ဥပမာ- \$x = 1;

```
while ($x < 6) {  
    $x++;  
    if ($x == 3) continue;  
    echo $x;  
}
```

⊛ while loop သည် endwhile statement သည် အတိတ်ကုန်ဆုံးသို့ ရောက်ရှိသည်။

ဥပမာ- \$x = 1;

```
while ($x < 6) {  
    echo $x;  
    $x++;  
} endwhile;
```

do-while loop

⊛ condition သည် check အတိတ်ကုန်ဆုံးသို့ ရောက်ရှိသည့်အခါမှ စတင်ပြီး condition သည်
check အတိတ်ကုန်ဆုံးသို့ ရောက်ရှိသည့်အခါမှ စတင်ပြီး condition သည်
အတိတ်ကုန်ဆုံးသို့ ရောက်ရှိသည့်အခါမှ စတင်ပြီး condition သည်

ဥပမာ- \$x = 1;

```
do {  
    echo $x;  
    $x++;  
} while ($x < 6);
```

⊛ အတိတ်ကုန်ဆုံးသို့ ရောက်ရှိသည့်အခါမှ စတင်ပြီး while loop သည် break; နှင့် continue; သို့မဟုတ်
အတိတ်ကုန်ဆုံးသို့ ရောက်ရှိသည်။

for loop

- ① loop එක ක්‍රියාකාරී කර ගන්න තිබේනම් ඉන්න එක; for loops සෙයිනු ඇත. ඔහු syntax එක ඉගෙනගන්න තිබේ.

```
for (expression1, expression2, expression3) {  
    // Code block  
}
```

expression 1 $\Rightarrow$ evaluated once

expression 2 $\Rightarrow$ evaluated before each iteration

expression 3 $\Rightarrow$ evaluated after each iteration

උදා: 1) 0 සිට 10 දක්වා අංකයන් output කරයි.

```
for ($x = 0; $x <= 10; $x++) {  
    echo "The number is: $x <br>";  
}
```

- ② මේ සඳහා ද, ඔබ ජර්නි break සහ continue statements සෙයිනු ඇත.

foreach loop

- ① array එකක සහ object එකක දී සියලුම properties ඉගෙනගන්න එක ජර්නි loop තිබේනම් ඔබ සෙයිනු ඇත.

- ② Array එකක් සඳහා foreach loop එකක් ඔබ ජර්නි සෙයිනු ඇත.

```
උදා: $colors = array("red", "blue", "green");  
foreach ($colors as $x) {  
    echo "$x <br>";  
}
```

- ③ ඔබ ඉගෙනගන්න Array එකක් 'indexed array' ලෙස හඳුන්වනු ලැබේ; 'associative array' (keys සහ values ලෙසට ඉගෙනගන්න) සඳහා ද, ඔබ ජර්නි foreach loop සෙයිනු ඇත. මෙයින් keys සහ values ලෙසට output කළ හැක.

```
උදා: $members = array("Peter" => "35", "Ben" => "25");  
foreach ($members as $x => $y) {  
    echo "$x : $y <br>";  
}
```

- ④ Object එකක් සඳහා foreach loop එකක් ඔබ ජර්නි සෙයිනු ඇත.

```

class Car {
    public $color;
    public $model;
    public function __construct($color, $model){
        $this->color = $color;
        $this->model = $model;
    }
}

$myCar = new Car("red", "Volvo");
foreach($myCar as $x => $y){
    echo "$x: $y <br>";
}

```

④ මේ කළා ද, break and continue statements සෙයිනු දෙන්න.

⑤ array item වෙනස් reference වෙලා අර්ගය change වෙලාවට පත්ල නැවැත්වෙයි සෙයිනු දෙන්න. ඔබ්බේ foreach loop තුළ & character සෙයිනු දෙන්න.

```

$colors = array("red", "green", "blue", "yellow");
foreach ($colors as &$x) {
    if ($x == "blue") $x = "pink";
}
var_dump($colors);

```

⑥ endforeach statement සෙයිනු ද, foreach loops අවසන් වෙන්න.

```

foreach ($colors as $x) {
    echo "$x <br>";
}
endforeach;

```

Functions

⑦ PHP හි 1000 කට ඉක්මනින් built-in functions තිබෙයි.

⑧ කිසිදු script එක තුළට අනෙකුත් script එකකින් reuse වෙලාවට, code reuse වෙලාවට functions යොදා ගන්න.

⑨ function වෙනස් වෙලාවට function keyword සෙයිනු දෙන්න.

```

function myMessage() {
    echo "Hello World";
}

```

* function name වෙනස් character වෙනස් කොට underscore වෙනස් කරන්න පුළුවනි. function names, case-sensitive නෙවෙයි.

⑩ function වෙනස් call වෙලාවට ද, අවශ්‍ය වෙලාවට අර්ගය යොදා ගන්න.

```

myMessage();

```

④ functions arguments information pass କରିও arguments ফেরত দেয়।

উদা- function familyName(\$fname) {
 echo "\$fname Refsnes.
";
}
familyName("Praveen");
familyName("Kasun");
familyName("Julia");

④ ফাংশনগুলিতে arguments নির্দিষ্ট করা হয়, ফেরত দেয়। ফাংশনগুলিতে comma (,) ব্যবহার করে আর্গুমেন্ট পাস করা হয়।

④ যেখানে default argument values ফেরত দেয়।

উদা- function setHeight(\$minheight = 50) {
 echo "The height is : \$minheight
";
}
setHeight(350);
setHeight(); // will use the default value of 50
setHeight(135);

④ value থেকে return statement থেকে ফেরত দেয়।

উদা- function sum(\$x, \$y) {
 \$z = \$x + \$y;
 return \$z;
}
echo "5 + 10 = ". sum(5, 10). "
";
echo "7 + 13 = ". sum(7, 13). "
";
echo "2 + 4 = ". sum(2, 4);

Arrays

④ values নির্দিষ্ট করে variable থেকে store করে arrays ফেরত দেয়।

উদা- \$cars = array("Volvo", "BMW", "Toyota");

④ কয়েক array types আছে।

1) Indexed arrays - Arrays with a numeric index

2) Associative arrays - Arrays with named keys

3) Multidimensional arrays - Arrays containing one or more arrays

④ Array থেকে data type থেকে values পাঠানো করা যায়।
array থেকে value নেওয়া যায়, বিভিন্ন data types থেকে values পাঠানো যায়।

উদা- \$myArr = array("Volvo", 15, ["apples", "bananas"], myFunc);

★ Array elements ମଧ୍ୟ elements ବଦଳାଇବା count () on-build function ଦ୍ଵାରା କରାଯାଇଥାଏ।

```
PHP:- $cars = array("BMW", "Toyota", "Ford");  
echo count($cars);
```

★ Indexed array elements ପ୍ରତି access କରିବା indexing ବ୍ୟବହାର କରାଯାଏ।

```
PHP:- echo $cars[0];
```

★ element ବଦଳାଇବା update କରିବା, ଏହା ମଧ୍ୟ କରାଯାଇଥାଏ।

```
PHP:- $cars[1] = "Ford";
```

★ array ବଦଳାଇବା ଏବଂ ନୂଆ element ବଦଳାଇବା କରିବା array-push() on-build function ବ୍ୟବହାର କରାଯାଏ।

```
PHP:- array_push($cars, "Ford");
```

★ Associative array elements ପ୍ରତି access କରିବା key name ବ୍ୟବହାର କରାଯାଏ।

```
PHP:- $car = array("brand" => "Ford", "model" => "Mustang");  
echo $car["model"];
```

★ element ବଦଳାଇବା update କରିବା, ଏହା ମଧ୍ୟ କରାଯାଇଥାଏ।

```
PHP:- $car["model"] = "Falcon";
```

★ Array ବଦଳାଇବା create କରିବା ଏହା array() on-build function ବ୍ୟବହାର କରାଯାଏ square brackets ([]) ବ୍ୟବହାର କରାଯାଏ।

```
PHP:- $cars = array("Volvo", "BMW", "Ford");  
$cars = ["Volvo", "BMW", "Ford"];
```

★ element ବଦଳାଇବା ଏହା function ବଦଳାଇବା ମଧ୍ୟ କରିବା, ଏହା access କରିବା parentheses () ବ୍ୟବହାର କରାଯାଏ।

```
PHP:- $myArr = array("Volvo", 15, myFunction);  
$myArr[2]();
```

★ Array ବଦଳାଇବା elements ବଦଳାଇବା କରିବା array-splice() on-build function ବ୍ୟବହାର କରାଯାଏ।

```
PHP:- $cars = array("Volvo", "BMW", "Toyota");  
array_splice($cars, 1, 1);
```

* ଏହା ଏକ ସ୍ଥାନରେ କେତେକ index ବଦଳାଇବା ମଧ୍ୟ କରାଯାଇଥାଏ, ଏହା number ବଦଳାଇବା କରିବା ଏହା delete କରାଯାଇଥାଏ elements ବଦଳାଇବା କରାଯାଏ।

* ଏହା ଏକ ସ୍ଥାନରେ କେତେକ elements ବଦଳାଇବା indexes array ବଦଳାଇବା normal indexing ବ୍ୟବହାର କରାଯାଏ।

★ elements delete କରିବା unset() on-build function ବ୍ୟବହାର କରାଯାଏ, ଏହା କେତେକ elements indexes ବଦଳାଇବା କରିବା କରାଯାଏ।

```

$cars = array("Volvo", "BMW", "Toyota");
unset($cars[1]);

```

- ⊛ Array element remove element remove function array_pop() on-build function array_shift() on-build function array_remove()

```

$cars = array_pop($cars);
array_shift($cars);

```

- ⊛ Array elements sort (sort) on-build functions

sort() - sort arrays in ascending order

rsort() - sort arrays in descending order

asort() - sort associative arrays in ascending order, according to the value

ksort() - sort associative arrays in ascending order, according to the key

arsort() - sort associative arrays in descending order, according to the value

krsort() - sort associative arrays in descending order, according to the key

```

1) $cars = array("Volvo", "BMW", "Toyota");

```

```

$numbers = array(4, 6, 2, 22, 11);

```

```

sort($cars);

```

```

rsort($numbers);

```

```

sort($numbers);

```

```

rsort($cars);

```

```

2) $age = array("Peter" => "35", "Ben" => "37", "Joe" => "43");

```

```

asort($age);

```

```

ksort($age);

```

```

arsort($age);

```

```

krsort($age);

```

Date and Time

- ⊛ date formatting function date() on-build function

- ⊛ date syntax

```

date(format, timestamp)

```

- * format - Required. Specifies the format of the timestamp
- * timestamp - Optional. Specifies a timestamp. Default is the current date and time.

⊛ ઝાન characters, dates અણો છુટો લેણો લેલો.

- d - represents the day of the month (01 to 31)
- m - represents a month (01 to 12)
- Y - represents a year (in four digits)
- l - (lowercase 'L') - represents the day of the week

જુઓ <?php

```
echo "Today is ". date("Y/m/d"). "<br>";
echo "Today is ". date("Y.m.d"). "<br>";
echo "Today is ". date("Y-m-d"). "<br>";
echo "Today is ". date("l");
?>
```

⊛ Time ડેટા output કરીને, date() લેલો જાણ. ઝાન characters લેણો લેલો.

- H - 24-hour format of an hour (00 to 23)
- h - 12-hour format of an hour with leading zeros (01 to 12)
- i - minutes with leading zeros (00 to 59)
- S - seconds with leading zeros (00 to 59)
- a - lowercase ante meridiem and post meridiem (am or pm)

જુઓ <?php

```
echo "The time is ". date("h:i:sa");
?>
```

⊛ હિલક ઝા ટીમો એમ જાણીને mktime() on-build function ડેટા, લેલો જાણ. એક હિલક ઝાન unix timestamp લેણો લેલો.

જુઓ ~~mktime~~

<?php

```
$d = mktime(11, 14, 54, 8, 12, 2024);
echo "Created date is ". date("Y-m-d h:i:sa", $d);
?>
```


classes and objects

⊛ classes define class keyword use code.

Ex:- <?php

```
class Fruit {  
    // code goes here  
}
```

⊛ class define properties define code, class property define set code (setting) and get code (getting) methods code define.

Ex:- <?php

```
class Fruit {  
    //properties  
    public $name;  
    public $color;  
  
    //Methods  
    function set_name($name){  
        $this->name = $name;  
    }  
    function get_name(){  
        return $this->name;  
    }  
}
```

⊛ objects create new keyword define.

<?php

```
class Fruit {  
    public $name;  
  
    function set_name($name){  
        $this->name = $name;  
    }  
    function get_name(){  
        return $this->name;  
    }  
}  
  
$apple = new Fruit();  
$banana = new Fruit();  
$apple->set_name('Apple');  
$banana->set_name('Banana');  
  
echo $apple->get_name();  
?>
```

★ color property එක සම්බන්ධව ජාති ප්‍රදර්ශනය කළහොත්.

උදා:- <?php

```
class Fruit {  
    public $color;  
    public $name;  
  
    function set_color($color) {  
        $this->color = $color;  
    }  
    function get_color() {  
        return $this->color;  
    }  
}  
  
$apple = new Fruit();  
$apple->set_color('Red');  
echo "Color: " . $apple->get_color();  
?>
```

★ මෙහි `$this` keyword එක සෙල්ලි ආදායක methods තුළ දී එමගින් මත පැවරුණු object (current object) එක පෙන්වයි.

★ කිසිකම object එකක් යම් class එකකට ප්‍රතිරෝධීතාව ඇති බව සනාථ කිරීමට `instanceof` keyword එක යොදා ගනී. `true` හෝ `false` ලෙස output එවයි.

උදා:- <?php

```
$apple = new Fruit();  
var_dump($apple instanceof Fruit); // outputs bool(true)  
?>
```